08. Model Evaluation

Evaluate cluster geometry, stability and profile separation—not classification accuracy.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

Reported K selection

The elbow analysis uses K-Means inertia for K=1..10 and KneeLocator returns K=4. Silhouette scores are computed and plotted for K=2..10, but the saved notebook does not print the numeric scores. Do not invent or quote a silhouette value.

Final cluster profiles

Metric (mean unless rate)	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Income	39,681	72,808	36,960	70,723
Total_Spending	222	1,237	166	1,190
NumWebPurchases	3.15	5.69	2.71	5.79
NumCatalogPurchases	0.97	5.50	0.84	5.01
NumStorePurchases	4.14	8.66	3.62	8.43
NumWebVisitsMonth	6.31	3.58	6.66	3.73
Response rate	7.62%	16.67%	14.19%	32.01%
Age	55.67	59.49	55.69	58.93
Total_Children	1.24	0.51	1.27	0.46

Living arrangement pattern

Cluster	Living_With_Alone	Living_With_Partner
0	0%	100%
1	0%	100%
2	99.3%	0.7%
3	100%	0%

Interpretation

- Clusters 0 and 2 have lower average income and substantially lower total spending than clusters 1 and 3.
- Clusters 1 and 3 have higher average income, more purchases across web/catalogue/store channels and fewer web visits per month.
- Clusters 0 and 1 are almost entirely coded as Partner; clusters 2 and 3 are almost entirely Alone. This household split appears strongly represented in the final solution.
- Cluster 3 has the highest observed Response rate (32.0%), while cluster 0 has the lowest (7.6%). This is descriptive and may partly reflect Response being included in the clustering inputs.

Cautions about the table

The notebook appends Agglomerative cluster labels to X, which is the original unscaled 18-feature encoded dataframe, then calculates group means. These means are interpretable in the original feature units. Cluster IDs 0–3 are arbitrary identifiers; they do not imply rank or order. The profile values describe this dataset and are not guaranteed to recur in future cohorts.

Evaluation not present

- No numeric silhouette table is printed.
- No external ground-truth segment labels exist, so classification metrics such as accuracy, precision and recall do not apply.
- No bootstrap or time-based stability analysis is shown.
- No formal business outcome test is reported.